

Part A Chapter 3

Bacterial Decision Making: Food Uptake by *E. coli*¹

1 Introduction

When a human or other organism performs an action, this is often based on a choice that was made between different possible actions. To make a choice between different possible options, in one way or the other a decision making process is performed. Decision making as performed by humans or other organisms is a wide area of research, involving, for example, conducting experiments, designing decision models, performing simulations, and investigating underlying (physical, biological, neurological or other) mechanisms for a given organism. Although brain imaging techniques are becoming more and more sophisticated, often they only give indications of the rough areas (in space and time) within which activations take place, not exactly which neurons are involved, and what is their specific temporal pattern of activation. Therefore, modelling decision making on the basis of these underlying brain processes is not feasible in detail (yet). For simpler, unicellular organisms such as the bacterium *E. coli*, the internal (chemical) processes are sufficiently accessible to obtain a decision model of, for example, their food import behaviour which is grounded in the underlying biochemistry; a source for this is, for example (Neidhardt, Curtiss III, Ingraham, Lin, Brooks Low, Magasanik, Reznikoff, Riley, Schaechter & Umberger, 1996).



1.1 Domain description and focus

To interact with its external environment in a useful manner, for *E. coli* various mechanisms are known. To get substances externally available inside, active import processes have to take place. *E. coli* has mechanisms to actively import specific food substances (of its choice), such as glucose or lactose; see for example (Widmaier, Raff, and Strang, 2006, pp. 113-119). As the action to import such a substance is to be initiated internally in the cell, it is crucial that it has information about the external world available inside. Imagine being in a closed box and

¹ Based on: Jonker, C.M., Snoep, J.L., Treur, J., Westerhoff, H.V., Wijngaards, W.C.A., BDI-Modelling of Complex Intracellular Dynamics. *Journal of Theoretical Biology*, vol. 251, 2008, pp. 1–23.

having to decide which things to take from the external environment, without seeing any of them. *E. coli* has mechanisms to find out which food substances are present in the environment: a form of sensing. In the simplified case considered here, only the decision making concerning glucose or lactose import is addressed. *E. coli* makes a decision between the two types of food in such a way that if lactose is externally present, but not glucose, then lactose is imported, and if glucose is externally present, then glucose is imported, but never lactose. For example, starting with an environment with lactose but not glucose, it will import lactose. However, if glucose is added to the environment (in addition to the lactose), then it will stop to import lactose and start to import glucose. This can be interpreted as: *E. coli* can use both substances, but it prefers glucose; it never imports both. How is *E. coli* realising this food uptake behaviour by its cellular processes? It is as if this organism through its internal processes makes decisions to put the two processes that are possible for the uptake of glucose or lactose 'on' or 'off' depending on the situation outside. These food uptake processes are in one way or the other controlled or regulated within *E. coli*.

Indeed all cells have facilities to regulate the processes within them. This takes place by producing specific types of proteins, called *enzymes* that enable certain chemical processes, or that speed up such processes. For each controlled process within the cell enzymes specific for that process are used; for example, to enable glucose uptake a specific glucose uptake enzyme is needed, and to enable lactose uptake a specific lactose uptake enzyme. So the question how *E. coli* controls the food uptake of glucose or lactose is now replaced by the question how the production of these specific enzymes is arranged. To produce enzymes each cell has a built-in process. In this chapter first a simple model is designed, where this production process for enzymes is modelled without taking into account the roles DNA and mRNA play. In Section 6 it will be discussed how these roles can be added to the model as well; see also (Widmaier, Raff, and Strang, 2006, pp. 61-86).

The enzyme for glucose import is (only) produced when a chemical substance called CRPcAMP is absent in the cell. The enzyme for lactose import is (only) produced when CRPcAMP is present in the cell, and when already some (low) concentration of lactose is present internally. The presence of the latter two substances relates to the presence of lactose and or glucose externally:

- when glucose is externally present, CRPcAMP is not internally present, and vice versa
- when lactose is externally present, then it is also internally present, and vice versa (a low amount of lactose automatically enters the cell by diffusion through the cell membrane)

Therefore, the internal presence of CRPcAMP and of lactose can be (and actually are) used as sensor information: indicators about the state of affairs of the external world.

1.2 Characteristic patterns in *E. Coli* food intake

The following characteristic patterns can be identified in the process described above. They can be used to evaluate whether the model shows relevant behaviour.

- If a sufficient amount of any food (lactose or glucose) is externally present, then a substantial amount of food will be internally present
- If food is present, then the bacterium will grow
- If both glucose and lactose are externally present, then glucose is imported but not lactose.

2 Analysis of the Main Aspects and Their Relations

In this section as a first step to design the model, the main concepts and their relations needed are identified.

2.1 Identification of relevant concepts

To describe the internal decision making processes taking care for the food uptake behaviour as discussed above, a number of concepts are needed. In the first place the *external presence* of lactose or glucose, and the *internal indicators* for this play a role. Furthermore, *enzymes* for the two types of uptake and the *uptake actions* themselves are relevant. Moreover, Finally, the effects of the import actions: *internal presence* of glucose or lactose and the *mass*. In summary, the following concepts play a role in the process:

External state

- glucose externally present
- lactose externally present

Internal states reflecting the external state

- CRPcAMP internally absent (glucose indicator)
- some lactose internally present (lactose indicator)

Internal states for action initiation

- enzyme for glucose import present
- enzyme for lactose import present

Action performance

- glucose import takes place
- lactose import takes place

Internal state showing the effect of actions

- substantial glucose internally present
- substantial lactose internally present
- mass

A concept is taken to occur when the related chemical concentration has crossed a particular threshold value. For example, glucose is externally present when glucose outside the cell has

a concentration of 0.1 mmol/l or more². The absence of CRPcAMP means that it has a concentration below a threshold of 0.01 mmol/l. The presence of some lactose internally means that the internal lactose concentration is elevated: above a threshold of 0.1 mmol/l. For initiated actions to be successful, enabling conditions in the environment need to be satisfied. For the import of lactose the enabling condition is the external presence of a sufficient level of lactose; for the import of glucose, of a sufficient level of glucose.

2.3 Relationships between the identified concepts

The dynamic relationships between the identified concepts are as follows. For a graphical overview, see Figure 1.

Sensory processes: from external state to internal indicators reflecting the external state

The dynamic relations for sensory processes describe that the external presence of glucose and or lactose affects the presence of the related internal indicators:

- glucose externally present affects CRPcAMP internally absent
- lactose externally present affects some lactose internally present

Action generation on the basis of internal indicators

The dynamic relations for action generation cover transcription (DNA affecting the presence of mRNA), translation (mRNA affecting the presence of enzyme), catalysis (enzymes affecting the related import reactions), and the effects of (co)factors in these steps.

- CRPcAMP internally absent affects enzyme for lactose import present
- some lactose internally present affects enzyme for lactose import present
- CRPcAMP internally absent affects enzyme for glucose import present
- enzyme for glucose import present affects that glucose import takes place
- enzyme for lactose import present affects that lactose import takes place

The effects of actions

The import actions affect that the related substance is internally present.

- glucose import taking place affects substantial glucose internally present
- lactose import taking place affects substantial lactose internally present
- glucose or lactose internally present affects mass

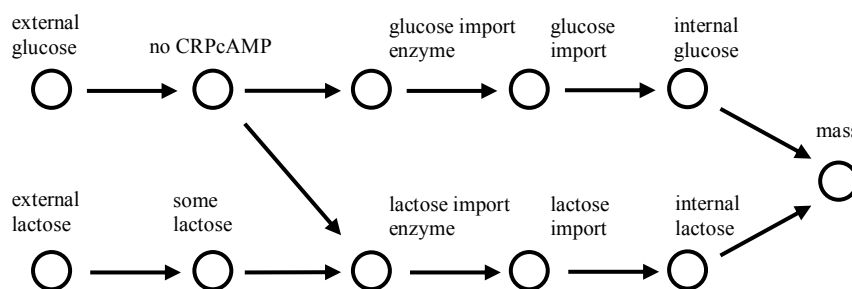


Figure 1 Dynamic relationships between the concepts

² mmol/l is milli-moles/liter; the mole is a standard SI (Système International) unit for counting particles such as molecules. One mole is about $6 \cdot 10^{23}$ molecules.

3 The Detailed Model

In this section more details of the model described here are discussed. First it is shown in which way the relevant concepts are formalised, and next the dynamic relations are formalised in a detailed manner.

3.1 Formalising the concepts used

To obtain a detailed model, first the concepts used are formalised. In this case for each of the concepts a numerical variable is introduced that takes the numbers 0 and 1 as values, except the mass, which has real numbers as values; see Table 1.

External state	
glucose externally present	<i>EXTgluc</i>
lactose externally present	<i>EXTlac</i>
Internal state reflecting external state (indicators)	
CRPcAMP internally absent	<i>glucIND</i>
some lactose internally present	<i>lacIND</i>
Internal states for action initiation	
enzyme for glucose import present	<i>glucimpENZ</i>
enzyme for lactose import present	<i>lacimpENZ</i>
Action performance	
glucose import takes place	<i>glucIMP</i>
lactose import takes place	<i>lacIMP</i>
Internal state showing the effect of actions	
substantial glucose internally present	<i>INTgluc</i>
substantial lactose internally present	<i>INTlac</i>
mass	<i>M</i>

Table 1 Formalisation of the relevant concepts by numerical variables

3.2 Formalising the dynamic relationships

The dynamics of the relevant concepts involve mutual interactions between them that can be expressed by temporal relationships, which are formalised in more detail below. They describe the basic parts of the process as shown in Figures 1 and 2.

Sensory processes

The dynamic relations for sensory processes describe how the external presence of glucose and or lactose leads to the presence of the related internal indicators.

From external glucose to glucose indicator

$$glucIND(t+\Delta t) = EXTgluc(t)$$

This relation is based on chemical mechanisms known from the literature that show how the presence of glucose externally leads to the suppression of CRPcAMP.

From external lactose to lactose indicator

$$lacIND(t+\Delta t) = EXTlac(t)$$

This relation is based on diffusion through the cell membrane of (low amounts of) lactose.

Action generation

The dynamic relations for action generation describe the decision model for the action(s) to choose, based on the internal indicator information.

From glucose indicator to enzyme for glucose import

$$glucimpENZ(t+\Delta t) = glucIND(t)$$

This relation describes the process to produce the enzyme for glucose import. This is dependent on the presence of the internal glucose indicator (the absence of CRPcAMP).

From lactose and glucose indicator to enzyme for lactose import

$$lacimpENZ(t+\Delta t) = lacIND(t) * (1 - glucIND(t))$$

This relation describes the process to produce the enzyme for lactose import. This is dependent on the presence of both the lactose and glucose indicator.

From enzyme to glucose import

$$glucIMP(t+\Delta t) = glucimpENZ(t)$$

This relation describes how the presence of the glucose import enzyme makes the glucose import reaction work (by catalysis) .

From enzyme to lactose import

$$lacIMP(t+\Delta t) = lacimpENZ(t)$$

This relation describes how the presence of the lactose import enzyme makes the lactose import reaction work (by catalysis) .

The effects of the actions

The effects of the import actions are that the related substance will be internally present, and is used, among others, to grow.

Effect of glucose import taking place

$$INTgluc(t+\Delta t) = glucIMP(t) * EXTgluc(t)$$

This relation describes how the glucose import reaction achieves the effect that glucose is present internally, given it is sufficiently present externally.

Effect of lactose import taking place

$$INTlac(t+\Delta t) = lacIMP(t) * EXTlac(t)$$

This relation describes how the lactose import reaction achieves the effect that lactose is present internally, given it is sufficiently present externally.

Effect on mass

$$M(t+\Delta t) = M(t) + \gamma (INTgluc(t) + INTlac(t)) * M(t)$$

This relation describes how the food internally present leads to growth of the mass of the bacterium (or bacteria, if as part of the process of growing also cell division is taking place). The presence of the food is counted per unit of mass; the overall amount is therefore obtained

by multiplication with the mass. Moreover, only part of the food is used for growing, indicated by an efficiency factor γ .

t	EXT gluc	EXT lact	gluc IND	lact IND	gluc ENZ	lact ENZ	gluc IMP	lact IMP	INT gluc	INT lact
0	0	1	0	0	0	0	0	0	0	0
1	0	1	1	1	0	0	0	0	0	0
2	0	1	1	1	0	0	0	0	0	0
3	0	1	1	1	0	1	0	0	0	0
4	0	1	1	1	0	1	0	1	0	0
5	0	1	1	1	0	1	0	1	0	1

Table 2 Dependencies over time for import actions

In Table 2 the dependencies over time are shown for an example scenario leading to lactose import. Table 3 shows these dependencies for growth.

t	int gluc	int lact	mass	add mass	γ
0	1	0	1	0.20	0.2
1	0	1	1.2	0.24	0.2
2	0	1	1.44	0.29	0.2
3	0	1	1.73	0.35	0.2
4	0	1	2.08	0.42	0.2
5	0	1	2.50	0.50	0.2

Table 3 Dependencies over time for growth

4 Simulation Experiments

The model as described above has been used to make a number of simulations in Microsoft Excel. As an example scenario an environment has been offered where sometimes lactose is present, sometimes glucose, and sometimes both, as shown in Figure 2, left hand column, upper two graphs. Notice that also a short period occurs where no food is present externally.

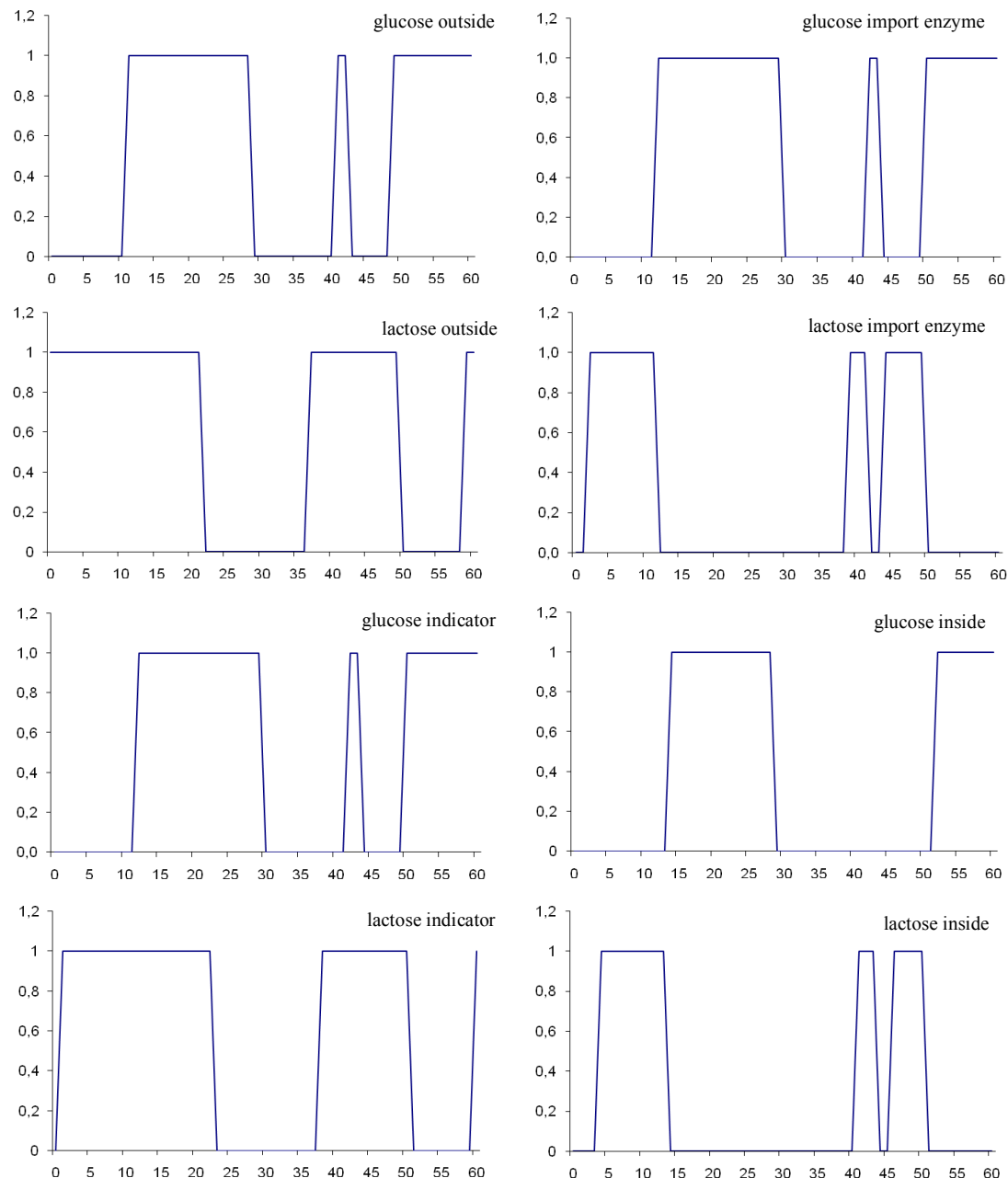


Figure 2 Example simulation trace: from external substances to internal substances

The related indicators internally in the cell are shown in the left hand column as well. In the right hand column of Figure 2, the decision process is shown, based on the information from

the indicators. Indeed, first the step to prepare lactose import is taken, whereas after the appearance of glucose in the environment lactose import preparation is withdrawn and instead the step to prepare import glucose is taken. In Figure 2 it is also shown how the enzymes lead to effects of having lactose or glucose inside. Notice that at one point in time no food uptake is performed whereas food is available externally. This happens because the bacterium is not prepared for the right type of food at that point in time. In Figure 3 the growth pattern is shown. In the periods without any food source the growth is stopped. Further, the growth pattern appears as exponential

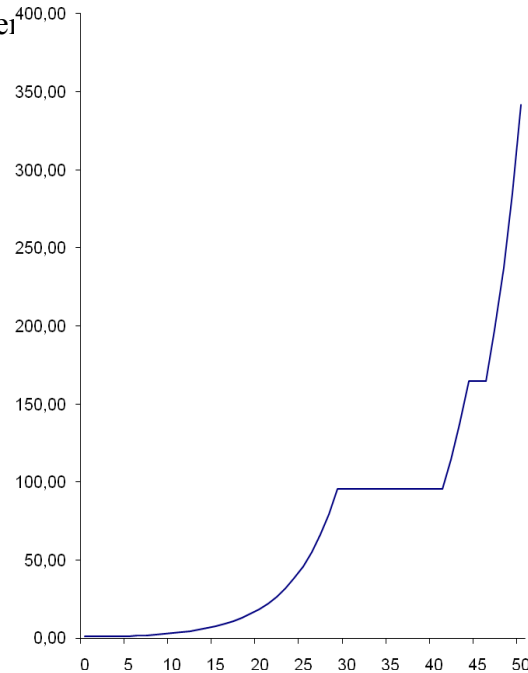


Figure 3 Example simulation trace: growth

5 Evaluation

In this section, first the simulations performed using the model are evaluated with respect to the characteristic patterns identified in Section 1.2. Next, the assumptions behind the model are summarised. Finally some possibilities of modification or refinement of the model are discussed.

5.1 Evaluation of simulations and characteristic patterns

A number of different simulations conducted in Microsoft Excel show all of the characteristic patterns identified in Section 1.2. It is not difficult to perform some more experiments under certain scenarios for the presence of food the environment.

5.2 Assumptions behind the model

The assumptions that have been made include:

- The dynamic relations specified are based on knowledge about the cell's biochemistry from the literature. This knowledge is assumed to be valid.
- The environment does not change too fast, so that *E. coli* has enough time to react on changes and is prepared for the right type of food.
- Each of the concepts can be represented in a binary manner by values 0 and 1.
- All reactions take the same amount of time Δt .
- A fixed fraction of the incoming food is used to grow.

5.3 Possibilities for variation and refinement

A number of possible variations or refinements can be made.

- In how far can the bacterium still grow when the circumstances in the environment fluctuate very fast.
- Incorporate the role of DNA and mRNA in the enzyme production process (for more details, see Section 6 below).
- Real numbers can be used instead of binary values.
- Different time differences for different processes.
- Explicit treatment of cell division as soon as a cell has grown enough.

6 Further Explorations

In this section it is discussed how the roles of DNA and mRNA can be explored further and added to the model.

6.1 The roles of DNA and mRNA: transcription and translation

The production process for enzymes in a cell goes via DNA and mRNA; see also (Widmaier, Raff, and Strang, 2006, pp. 61-86), and Figure 4.

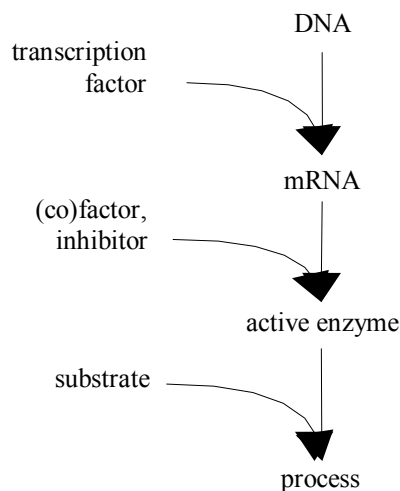


Figure 4 Generic control structure for processes in a cell

Before a specific active enzyme is produced, first for that enzyme a specific messenger RNA, mRNA is to be produced. So, for example, to produce the lactose import enzyme, mRNA for lactose import is needed. These specific types of mRNA are produced on the basis of specific locations at the cell's DNA: operons. If the cell's DNA has an operon for lactose import, then under certain conditions (the presence or absence of some other substances, called transcription factors) the cell will produce mRNA for lactose import. These conditions may relate to states within the cell that reflect the circumstances outside, for example the fact that lactose is available in the environment. So, going through the different steps involved:

- *Creating internal states that reflect the state of the environment*
Circumstances in the external environment lead to certain concentrations of specific internal substances reflecting the external state of affairs, for example, that lactose is present outside.
- *From DNA to mRNA*
On the basis of specific operons on the available DNA, and some of the substances within the cell that reflect the state within and outside the cell (called transcription factors), a specific mRNA is produced; for example mRNA for lactose import is produced (this process is sometimes called *transcriptional regulation*).
- *From mRNA to enzyme*
On the basis of a specific available mRNA, and certain specific internal substances that reflect the state within and outside the cell (called cofactors or inhibitors) a specific active enzyme is produced; for example from the lactose import mRNA, the lactose import enzyme is produced (this process is sometimes called *translational regulation*).
- *From active enzyme to process*
The specific enzymes enable the corresponding cell's process, assuming that the substrate needed for this process is available (this is sometimes called *catalysis*). Such processes can also produce substances that are used in the previous steps as transcription factors, cofactors or inhibitors, thus closing the control loop.

Notice that for this whole process to work as an adequate control system it is crucial that each of these steps described above (see also Figure 5) can be affected by other substances in the cell that can block or facilitate it (transcription factors, cofactors, inhibitors); see, for example, (Widmaier, Raff, and Strang, 2006, p. 70, Table 3-3). For the uptake of lactose and glucose, *E. coli*'s DNA includes two specific parts (operons). The operon for glucose import is always transcribed into the relating mRNA for glucose import. The operon for lactose import is (only) transcribed into mRNA for lactose import when a chemical substance called CRPcAMP is present in the cell, and when already some (low) concentration of lactose is

present internally. The presence of the latter two substances relates to the presence of lactose and or glucose externally, as discussed in Section 1 above. Therefore, the internal presence of CRPcAMP and of lactose are used as sensor information: indicators about the state of affairs of the external world.

6.2 Analysis of the Main Aspects and Their Relations

In this section the main concepts and their relations for DNA and mRNA are identified.

6.2.1 Identification of relevant concepts

Additional concepts are the *DNA operons* for lactose uptake and for glucose uptake, and the corresponding types of *mRNA* are added.

Internal states preparing for actions

- DNA operon for glucose import present
- DNA operon for lactose import present
- mRNA for glucose import present
- mRNA for lactose import present

6.2.2 Relationships between the identified concepts

The dynamic relationships involving the identified additional concepts are as follows. For a graphical overview, see Figure 5.

Action generation on the basis of internal indicators

The dynamic relations for action generation cover transcription (DNA affecting the presence of mRNA), translation (mRNA affecting the presence of enzyme), catalysis (enzymes affecting the related import reactions), and the effects of (co)factors in these steps.

- DNA operon for glucose import present affects mRNA for glucose import present
- DNA operon for lactose import present affects mRNA for lactose import present
- CRPcAMP internally absent affects mRNA for lactose import present
- some lactose internally present affects mRNA for lactose import present
- mRNA for glucose import present affects enzyme for glucose import present
- CRPcAMP internally absent affects enzyme for glucose import present
- mRNA for lactose import present affects enzyme for lactose import present
- some lactose internally present affects enzyme for lactose import present
- enzyme for glucose import present affects that glucose import takes place
- enzyme for lactose import present affects that lactose import takes place

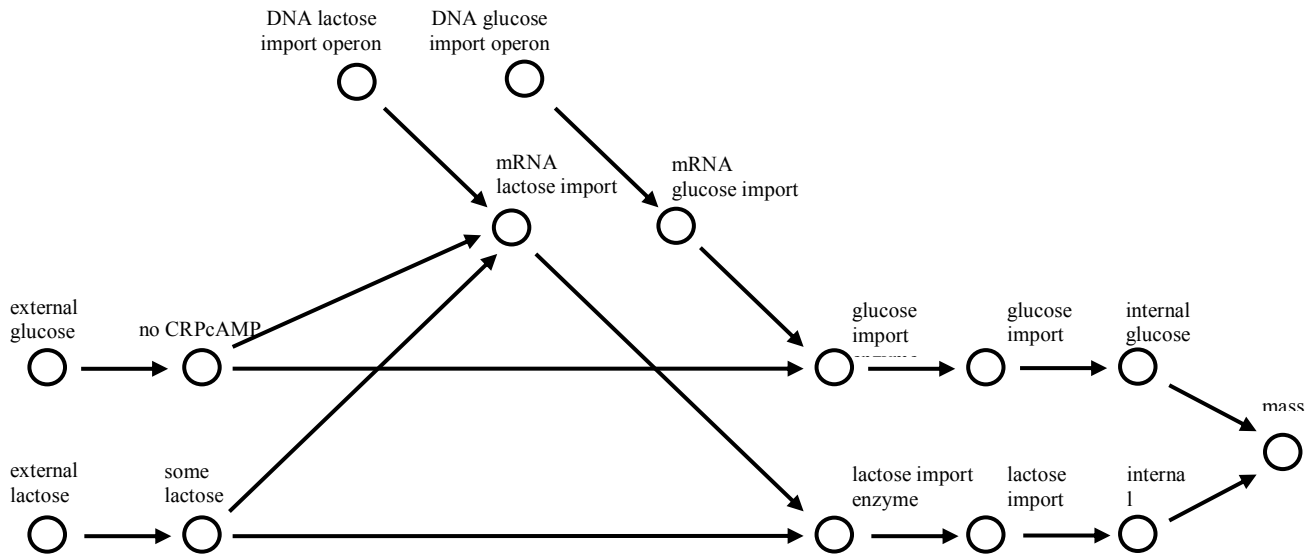


Figure 5 Dynamic relationships between the concepts

6.3 The Detailed Model

In this section more details of the model described here are discussed. First it is shown in which way the relevant concepts are formalised, and next the dynamic relations are formalised in a detailed manner.

6.3.1 Formalising the additional concepts used

To obtain a detailed model, first the concepts used are formalised. In this case for each of the concepts a numerical variable is introduced that takes the numbers 0 and 1 as values, except the mass, which has real numbers as values.

Internal state preparing for actions

- DNA operon for glucose import present *DNAGluc*
- DNA operon for lactose import present *DNALac*
- mRNA for glucose import present *mRNAGluc*
- mRNA for lactose import present *mRNALac*

6.3.2 Formalising the dynamic relationships

The dynamics of the relevant concepts involving action generation are formalised in more detail below. They describe the basic parts of the process as shown in Figure 6. The other parts of the process are as described in Section 3.

Action generation

The dynamic relations for action generation describe the decision model for the action(s) to choose, based on the internal indicator information. This decision process covers transcription

(from DNA to mRNA), translation (from mRNA to enzyme), catalysis (enzymes affecting the related import reactions), and the (co)factors enabling and/or blocking these steps.

From DNA operon to mRNA for glucose import

$$mRNA_{gluc}(t+\Delta t) = DNA_{gluc}(t)$$

This relation describes the transcription of the DNA operon for glucose import to obtain the corresponding mRNA. This takes place unconditionally.

From DNA operon to mRNA for lactose import

$$mRNA_{lac}(t+\Delta t) = DNA_{lac}(t) * (1 - glucIND(t)) * lacIND(t)$$

This relation describes the transcription of the DNA operon for lactose import to obtain the corresponding mRNA. This depends on the presence of the internal lactose indicator (presence of some lactose internally) and on the presence of the internal glucose indicator (the absence of CRPcAMP internally).

From mRNA to enzyme for glucose import

$$glucimpENZ(t+\Delta t) = mRNA_{gluc}(t) * glucIND(t)$$

This relation describes the translation process to produce the enzyme for glucose import. This is dependent on the presence of the internal glucose indicator (the absence of CRPcAMP).

From mRNA to enzyme for lactose import

$$lacimpENZ(t+\Delta t) = mRNA_{lac}(t) * lacIND(t)$$

This relation describes the translation process to produce the enzyme for lactose import. This is dependent on the presence of the lactose indicator.

From enzyme to glucose import

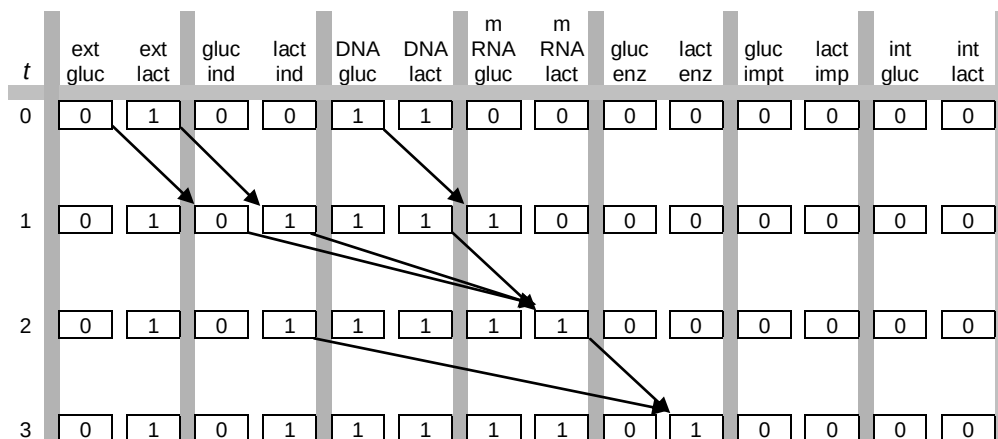
$$glucIMP(t+\Delta t) = glucimpENZ(t)$$

This relation describes how the presence of the glucose import enzyme makes the glucose import reaction work (by catalysis) .

From enzyme to lactose import

$$lacIMP(t+\Delta t) = lacimpENZ(t)$$

This relation describes how the presence of the lactose import enzyme makes the lactose import reaction work (by catalysis).



4	0	1	0	1	1	1	1	1	0	1	0	1	0	0
5	0	1	0	1	1	1	1	1	0	1	0	1	0	1

Table 4 Dependencies over time for import actions

In Table 4 the dependencies over time are shown for an example scenario leading to lactose import. Table 5 shows these dependencies for growth.

t	int gluc	int lact	mass	add mass	γ
0	1	0	1	0.20	0.2
1	0	1	1.2	0.24	0.2
2	0	1	1.44	0.29	0.2
3	0	1	1.73	0.35	0.2
4	0	1	2.08	0.42	0.2
5	0	1	2.50	0.50	0.2

Table 5 Dependencies over time for growth

6.4 Simulation Experiments

The model as described above has been used to make a number of simulations in Microsoft Excel. As an example scenario an environment has been offered where sometimes lactose is present, sometimes glucose, and sometimes both, as shown in Figure 6, left hand column, upper two graphs. A short period occurs where no food is present externally.

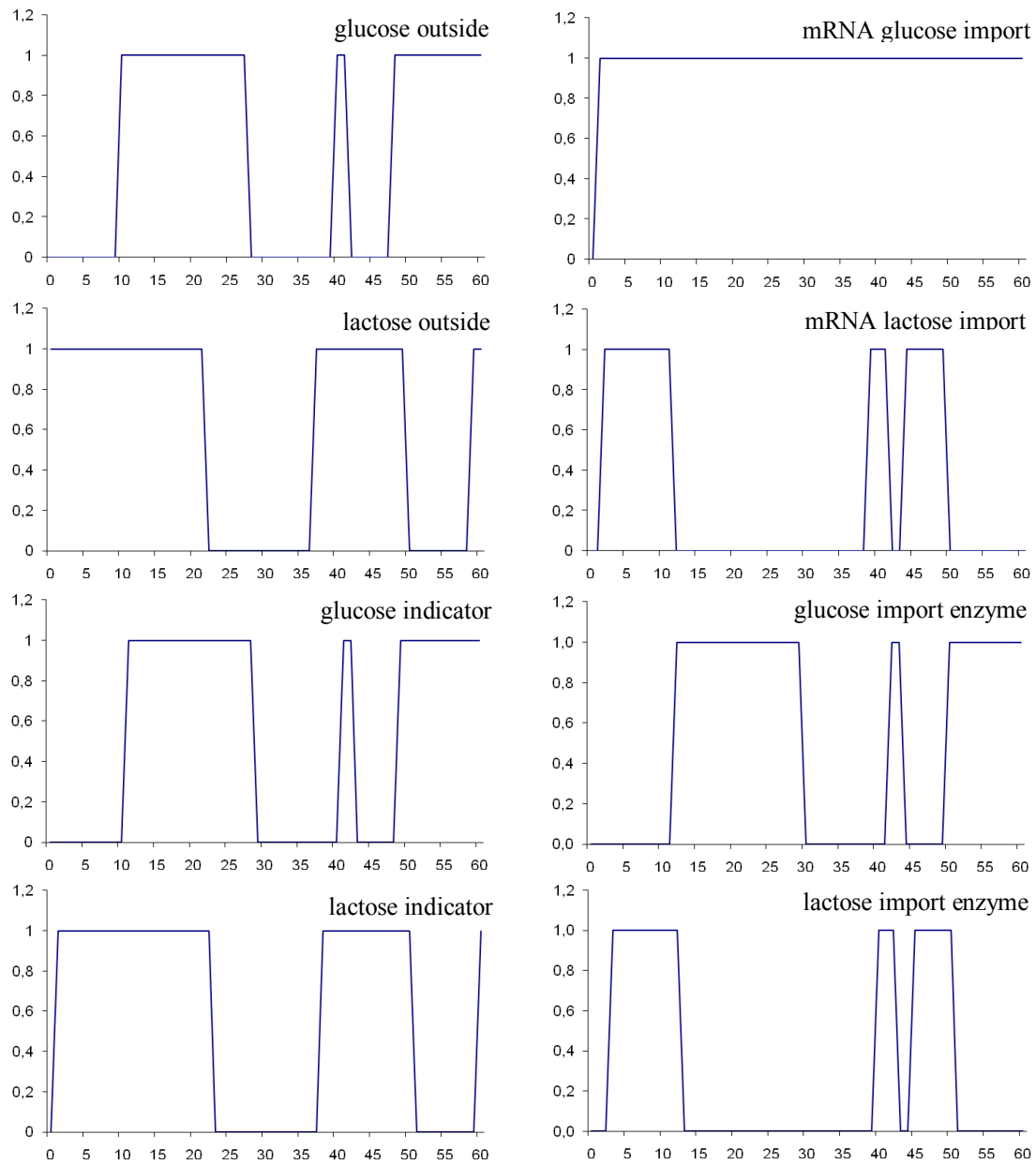


Figure 6 Example simulation trace: from external substances to controlling enzymes

The related indicators internally in the cell are shown in the left hand column as well. In the right hand column of Figure 6, the decision process is shown, based on the information from the indicators. Indeed, first all steps to prepare lactose import are successfully taken, whereas after the appearance of glucose in the environment these lactose import preparation steps are withdrawn and instead all steps to prepare import glucose are taken.

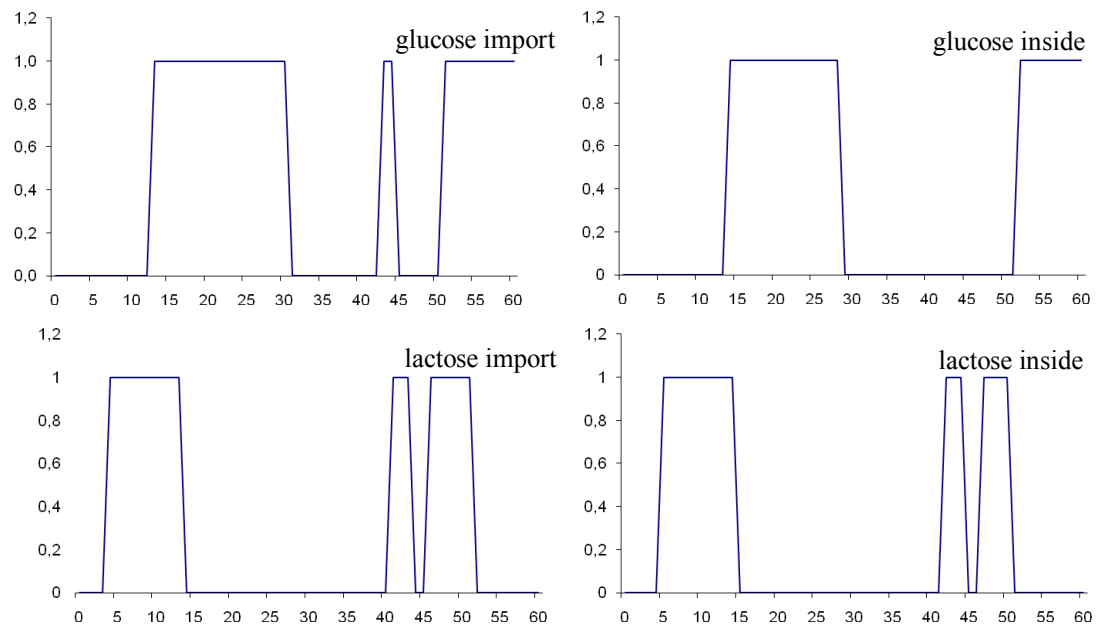


Figure 7 Example simulation trace: from import actions to internal substances

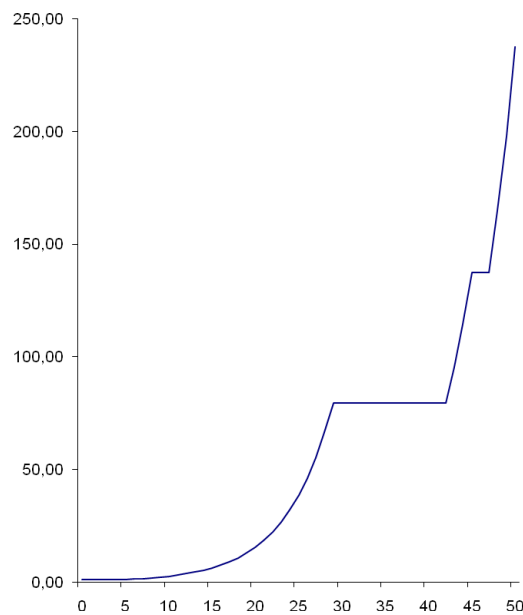


Figure 8 Example simulation trace: bacterial growth

In Figure 7 it is shown how the enzymes lead to the corresponding import actions and the effects of having lactose or glucose inside. This happens because the bacterium is not prepared for the right type of food at that point in time. In Figure 8 the growth pattern is shown. In the periods without any food source the growth is stopped. Further, the growth pattern appears as exponential, as can be checked in the graph.

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